作业纸

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6.4 (1) 振动周期 $T=\frac{2\pi}{\omega}$，$\omega=\frac{v_m}{A}=1.5\ rad/s$　解 $T=4.19s$

(2) $a_m=v_m\cdot\omega=4.5\ cm/s^2$

(3) $x=A\cos(\omega t+\varphi)$

$t=0$ 时，v 达到最大，x 最小 则 $\varphi=-\frac{\pi}{2}$

$x=2\sin\frac{3}{2}t$ (cm)

$=2\cos(\frac{3}{2}t-\frac{\pi}{2})$ (cm)

6.8 (1) 总能量 $E=E_k+E_p=0.8J$

$E=\frac{1}{2}kA^2$

解 $A=0.25m=25cm$

(2) $\frac{1}{2}kx^2=\frac{E}{2}$

解 $x=17.7cm$

(3) $E_p=\frac{1}{2}k(\frac{A}{2})^2=0.2J$

6.11 串联后弹簧的劲度系数 k 有 $\frac{1}{k}=\frac{1}{k_1}+\frac{1}{k_2}$

即 $k=\frac{k_1k_2}{k_1+k_2}$

其振动周期 $T=2\pi\sqrt{\frac{m}{k}}=2\pi\sqrt{\frac{(k_1+k_2)m}{k_1k_2}}$

6.14 (1) $F=\frac{GM_E'm}{r^2}=\frac{GM_Em}{r^2}\cdot\frac{r^3}{R_E^3}=\frac{GM_Emr}{R_E^3}$

方向指向地心

(2) $F'=-F\cdot\frac{x}{r}=-\frac{GM_Em}{R_E^3}x=-kx$

故该球做简谐运动

$T=2\pi\sqrt{\frac{m}{k}}=2\pi\sqrt{\frac{R_E^3}{GM_E}}$

(3) $\frac{GM_Em}{R_E^2}=m\frac{4\pi^2}{(T')^2}R_E$

解 $T'=2\pi R_E\sqrt{\frac{R_E}{GM_E}}$

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6.16

设木板沿水平方向移动了x

$$f_1 = \mu \frac{mg(l+x)}{l}$$

$$f_2 = \mu \frac{mg(l-x)}{l}$$

$F = f_2 - f_1$ 解 $F = -\frac{2\mu mg}{l}x$

故木板将作简谐运动

$$k = \frac{2\mu mg}{l} = \frac{mg}{l}$$

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{l}{g}} = 0.90s$$

6.22 $\frac{\theta_1}{\theta_0} = e^{-\beta\Delta t}$

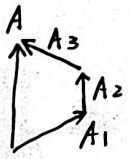
解 $\beta = \frac{1}{\Delta t}\ln\frac{\theta_0}{\theta_1} = 2.23\times10^{-3}\ s^{-1}$

$\tau = \frac{1}{2\beta}$

$Q = 2\pi\frac{\tau}{T}$

解 $Q = \frac{\pi}{\beta}\cdot\frac{1}{T} = \frac{1}{2\beta}\sqrt{\frac{g}{l}} = 702$

6.25 (1) 合振动的角频率不变 即 $\omega = 314\ rad/s$



振幅 $A = A_1\sin\frac{\pi}{6} + A_2\cdot\sin\frac{\pi}{2} + A_3\cdot\frac{\pi}{6}$

$= 0.16m$

~~初相~~ 由图可知 初相 $\varphi = \frac{\pi}{2}$

振动表达式 $x = 0.16\cos(314t + \frac{\pi}{2})$ (m)

(2) $A\cos(314\Delta t + \frac{\pi}{2}) = \frac{\sqrt{2}}{2}A$

$\Delta t = \frac{\frac{5}{4}\pi}{314}\ s = 1.25\times10^{-2}s$

6.26. $x = 0.06\cos(20\pi t)$

$y = 0.04\cos(20\pi t + \frac{\pi}{2}) = -0.04\sin(20\pi t)$

易知 $\left(\frac{x}{0.06}\right)^2 + \left(\frac{y}{-0.04}\right)^2 = \cos^2(20\pi t) + \sin^2(20\pi t) = 1$

$\frac{x^2}{0.06^2} + \frac{y^2}{0.04^2} = 1$

故轨迹为椭圆，质点绕行是右旋的

半长轴长为0.06 m，半短轴长为0.04 m

$T = \frac{2\pi}{\omega} = \frac{2\pi}{20\pi}\ s = 0.1s$

6.27 由题知 $\frac{V_x}{V_y} = \frac{3}{2}$，$V_x = 2.70\times10^4\ Hz$

解 $V_y = 1.80\times10^4\ Hz$

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